PROJECT SUTRA — 3-MEMBER GRAND FINALS PITCH

HARIKA (HOOK & PROBLEM) • VEDANTH (VISION & GCS) • NIKHIL (FLIGHT, COMMS & CLOSING)

TEAM: HARIKA • VEDANTH • NIKHIL
TOTAL TIME: ~2 MIN 15 SEC

⚡ SIMPLE VOCABULARY • 3-SPEAKER SPLIT: Plain English delivery with zero complicated jargon. Harika sets the problem reality, Vedanth presents the AI vision & ground control, and Nikhil commands the flight, comms, and final performance benchmarks.

SLIDE 01 Introduction: Project SUTRA

00:00 - 00:30 (30s)

SPEAKER 1: HARIKA (OPERATIONS LEAD) [STAGE: CENTER • HOOK & OVERVIEW]

> 🔥 **HOOK: "When over 300 lives were lost in the July 2024 Wayanad landslides, and 15,000 pilgrims were trapped in Kedarnath, rescue teams had only 72 hours to save lives. But the moment standard drones entered thick forests and mountain valleys, they lost signal and crashed."**

"Respected jury, we are **Team OFFGRID**. I am Harika, and with me are Vedanth and Nikhil. We built **Project SUTRA** — an autonomous drone swarm that finds survivors even when GPS fails and normal radios stop working."

- PILLAR 1 **Self-Flying Swarms:** Drones dodge trees and coordinate together at 50 times a second without ground towers. [Slide: 50Hz GNC]
- PILLAR 2 **Smart AI Video:** Live video stays connected through deep mountain valleys without blacking out. [Slide: -5dB JSCC]
- PILLAR 3 **Thermal & 3D AI Vision:** Spots survivors through smoke and finds their exact spot on steep slopes. [Slide: 3D DEM • 3.59cm]

SLIDE 02 The Problem: 4 Major Breakdowns

00:30 - 01:05 (35s)

SPEAKER 1: HARIKA [STAGE: URGENT & FACTUAL DISASTER AUDIT]

> ⚠️ **HOOK: "When we looked at real rescue reports from 2024, we found 4 major breakdowns:"**

- BREAKDOWN 1 **Trees Block GPS:** In Wayanad, thick tree cover blocked satellite signals, causing **70% of drones to crash into branches**.
- BREAKDOWN 2 **Mountains Block Radio:** In Kedarnath, mountain ridges blocked radio signals, dropping video feeds to total black.
- BREAKDOWN 3 **Slope Errors:** In Himachal and Sikkim, steep hillsides tricked standard drone cameras, **missing survivor locations by 15 to 30 meters**.
- BREAKDOWN 4 **Pilot Overload:** Flying 5 drones took **15 to 25 people** and 2 hours to set up, costing ₹12.5 Lakhs per flight.

HANDOFF TO VEDANTH → "Vedanth will now explain our AI vision and ground control solutions."

SLIDE 03 (A) AI Vision & Ground Control

01:05 - 01:35 (30s)

SPEAKER 2: VEDANTH (AI PERCEPTION & GCS LEAD) [STAGE: CONFIDENT & PRACTICAL]

"Thank you, Harika. I lead our AI perception and command systems. To solve the vision and control bottlenecks:"

- MOAT C **Thermal & 3D AI Vision:** Combines body heat sensors with 3D slope mapping on a Jetson Orin. It sees through smoke and locks onto survivors in under 15 milliseconds, **completely removing the 30-meter slope error**. [Slide: 3D DEM • 3.59cm Residual]
- MOAT D **1-Person Command Tablet:** An offline 3D tactical map allows **just 1 operator to fly the entire drone swarm**, bringing mission costs down from ₹12.5 Lakhs to just **₹95,000**. [Slide: Pegasus 3D GCS]

HANDOFF TO NIKHIL → "Now, our Tech Lead, Nikhil, will explain our autonomous flight, comms, and final performance benchmarks."

SLIDE 03 (B) & FINAL **Autonomous Flight, Comms & Benchmark ROI** 01:35 - 02:20 (45s)

SPEAKER 3: NIKHIL (TECH LEAD & SWARM ARCHITECT) [STAGE: TECHNICAL AUTHORITY & CLOSING PUNCH]

"Thank you, Vedanth. On the core flight and communications architecture:"

MOAT A **Self-Flying Swarms:** Drones calculate collision-free flight paths onboard at 50 times a second — zero ground towers, zero GPS reliance. [Slide: SUTRA-FSD & 3D ORCA]

MOAT B **Smart AI Video Link:** When signals drop behind mountain ridges, our neural link keeps the video running down to -5dB without blacking out. [Slide: Deep JSCC | -5dB]

"The bottom line: SUTRA searches a full square mile in just 10 minutes — 4 times faster than human rescue crews, saving lives before time runs out."

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi²	2-3 hours	45-60 min	10-18 min (4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000

"Thank you, respected jury. We are now open for your questions."